

Mingyi Wu

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Technical Skills

Proficient in designing and developing scalable data pipelines using Python, Java, and Apache Spark,

Strong background in data analysis and visualization using **Pandas, Tableau, and Matplotlib.**

Skilled in **RESTful API** development and **microservices architecture**, with hands-on experience in **Spring Boot, Node.js**, and **React.js**, leveraging **AWS** services such as S3, EMR, Lambda, and Redshift

Experienced in distributed data processing frameworks (**Hadoop, Spark, Kafka**) and relational and NoSQL databases (**MySQL, MongoDB**).

Familiar with automated testing tools (**Selenium, Postman**) to ensure high-quality software delivery.

Education

MSc in Professional Computer Science

Sep 2022 - Jan 2024

Simon Fraser University, Burnaby, Canada

GPA 3.75/4.00

Concentration in Big Data

Bachelor of Computer Science & Mathematics Honours

Dec 2018 - April 2022

Carleton University, Ottawa, Canada

GPA 3.72/4.00

Concentration in Computing Theory and Mathematics

Experience

Data Engineer

Jun 2024 – May 2025

Data Center of Guizhou Provincial Meteorological Service

- Cleaned and standardized over 300,000 multi-source datasets from meteorological stations using **Pandas, NumPy** and **Python**, to ensure data consistency and usability.
- Developed **RESTful APIs** using **Python** (FastAPI/Flask) to processed data to system platforms.
- Wrote **Java** programs to automate data process workflows, reducing manual effort by 85%
- Collaborated with researchers to Integrated quality-controlled data into model training workflows.

Full Stack Developer

Jan 2024 – May 2024

Guizhou Institute of Mountain Meteorology

- Developed and maintained **RESTful APIs** using **Spring Boot**, ensuring high performance and scalability.
- Developed front-end user interfaces using **React**, enhancing experience on browse-based applications.
- Optimized SQL databases by **PostgreSQL** to support data storage, complex querying and integration with back-end services, the response time was reduced by 27%.
- Performed Java Virtual Machine (JVM) tuning by detecting and solving memory leaks.

Research Assistant on Autonomous Vehicle Project

May 2023 – Dec 2023

Simon Fraser University, Department of Computer Science

- Extraction and analysis over 200,000 vehicle records using **Python** and **Pandas**.
- Designed and implemented a **MariaDB** database to integrate sequential driving frame data, enabling temporal analysis and visualizations with **Seaborn**.
- Implemented **Bayesian network** by **java-based** learning tools, yielding an interpretable model

Project Experience

ChatterSphere – Twitter Clone

Jun 2024 – Nov 2024

- Developed a full-stack social media platform using **React, Java Spring Boot**, integrated with **Google Cloud MySQL**, supporting 1000+ concurrent users and real-time interactions
- Designed and implemented RESTful APIs with Spring Boot to handle actions in the chat.
- Implemented **JWT-based** authentication and integrated **Cloudinary** for image management